GlobalTrade Logistics

How to Set Up and Run the Application

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This guide walks you through getting GlobalTrade running on your own computer, from a blank machine to a working login screen. You don't need to know anything about the code — just follow the steps in order.

Set aside about **30–45 minutes** the first time. Most of that is downloading and installing; the actual setup is short.

What's in this guide

1. What GlobalTrade is

GlobalTrade is a shipping and logistics system. It keeps track of cargo shipments moving between ports around the world — who booked them, which containers they use, which ship is carrying them, and where they've got to.

There are three kinds of user, and each sees a different part of the system:

- **Customers** book shipments and follow their progress.
- **Coordinators** manage those shipments day to day and move them along as they're loaded, sail, and arrive.
- **Administrators** look after the containers, the ships, the user accounts, and the system's activity records.

The application runs on your own computer. Nothing is sent anywhere online.

2. What you'll need to install

Four things, in this order. Each one is a free download.

What	Version	Why it's needed
Java (JDK)	17	The application is written in Java, so your computer needs Java to run it
MySQL Server	8.x	Stores all the shipment, ship, container and user information
Payara Server	6 (Community)	The web server that actually runs the application
Maven	3.8+	Assembles the application into a single installable file

> **Already have some of these?** That's fine, but check the versions. Java in particular must be **17** — newer versions such as 21 will cause problems. If you use IntelliJ IDEA, Maven is already bundled and you can skip installing it separately.

3. Step 1 — Install Java

1. Go to `https://adoptium.net/temurin/releases/?version=17` and download the JDK 17 installer for your operating

system.
2. Run the installer and accept the defaults.
3. Open a terminal (Command Prompt or PowerShell on Windows, Terminal on Mac or Linux) and type:

   ```
   java -version
   ```

   You should see something starting with `openjdk version "17`.

> **If you have more than one version of Java installed**, set the `JAVA_HOME` environment variable to point at the Java 17 folder. Everything else in this guide uses whichever Java that setting points to, so getting it wrong causes confusing errors later.

4. Step 2 — Install MySQL and load the data

MySQL is the database. It stores everything the application remembers.

1. Install MySQL Server 8 — use the MySQL Installer on Windows, `brew install mysql` on Mac, or your package manager on Linux.
2. During installation you'll be asked to set a password for the `root` user. **Write it down** — you'll need it in Step 4.
3. Check you can log in:

   ```
   mysql -u root -p
   ```

   Enter your password when prompted. Type `exit` to come back out.
4. Now load the starter data. From the project folder, run:

   ```
   mysql -u root -p < db/create_tables_and_seed_data.sql
   ```

   This creates the database and fills it with sample ports, ships, containers, shipments and user accounts so there's something to look at straight away.

> **This wipes any existing GlobalTrade data.** The script starts fresh every time it's run. That's deliberate — it means you can always get back to a known starting point — but don't run it again if you've entered data you want to keep.

To check it worked, log back in and run:

```
USE globaltrade;
SHOW TABLES;
SELECT COUNT(*) FROM users;
```

You should see seven tables listed and a count of **6** users.

5. Step 3 — Install the Payara server

Payara is the web server. It's what makes the application available in your browser.

1. Download Payara Server 6 (Community edition) from `https://www.payara.fish/downloads/`
2. Unzip it somewhere simple, such as `C:\payara6` on Windows or `~/payara6` on Mac or Linux. Avoid folder names with spaces.
3. Start it. Open a terminal, go into the `bin` folder, and run:

   ```
   cd payara6/bin
   asadmin start-domain
   ```

   On Mac or Linux, use `./asadmin start-domain`.
4. Wait a few seconds. When you see **Domain domain1 started**, it's running.
5. Check it by opening `http://localhost:4848` in your browser. This is Payara's control panel, which you'll use in the next step.

To stop the server later, run `asadmin stop-domain` from the same folder.

6. Step 4 — Connect the server to the database

Payara and MySQL are both running now, but they don't know about each other yet. This step introduces them. It's the fiddliest part of the setup, so take it slowly.

4a. Give Payara the MySQL connector

Payara needs a small file that lets it talk to MySQL. Find this file:

```
mysql-connector-j-8.4.0.jar
```

You can download it from `https://dev.mysql.com/downloads/connector/j/`, or if you've already built the project once, it will be in your Maven folder at `.m2/repository/com/mysql/mysql-connector-j/8.4.0/`.

Copy that file into:

```
payara6/glassfish/domains/domain1/lib/
```

Then restart Payara so it notices the new file:

```
asadmin restart-domain
```

Don't skip this. If you do, everything in the next part will look like it worked, but the connection test will fail with a message about a missing class. Copying this one file first saves a lot of head-scratching.

4b. Create the database connection

In the Payara control panel at `http://localhost:4848`, go to **Resources → JDBC → JDBC Connection Pools** and click **New**. Fill in:

Field	Value
Pool Name	`GlobalTradePool`
Resource Type	`javax.sql.DataSource`
Datasource Classname	`com.mysql.cj.jdbc.MysqlDataSource`

Click Next. On the following page, scroll down to **Additional Properties**. Delete any properties already listed there, then add these six:

Name	Value
`serverName`	`localhost`
`portNumber`	`3306`
`databaseName`	`globaltrade`
`user`	your MySQL username (usually `root`)
`password`	the MySQL password from Step 2
`useSSL`	`false`

Save it. Then open the pool again and click the **Ping** button at the top. If you see **Ping Succeeded**, Payara and MySQL are talking to each other and you can move on. If not, jump to the last section of this guide.

4c. Give the connection a name

Go to **Resources → JDBC → JDBC Resources** and click **New**:

Field	Value
JNDI Name	`jdbc/GlobalTradeDS`
Pool Name	`GlobalTradePool`

> **Type the name exactly as shown** — `jdbc/GlobalTradeDS`, capital letters included. The application looks for this exact name, and if it's even slightly different the application won't start.

Prefer the command line?

The same two things can be done with three commands instead. Replace `YOURPASSWORD` with your MySQL password:

```
asadmin create-jdbc-connection-pool --datasourceclassname com.mysql.cj.jdbc.MysqlDataSource --restype
javax.sql.DataSource --property
serverName=localhost:portNumber=3306:databaseName=globaltrade:user=root:password=YOURPASSWORD:useSSL=false
GlobalTradePool

asadmin ping-connection-pool GlobalTradePool

asadmin create-jdbc-resource --connectionpoolid GlobalTradePool jdbc/GlobalTradeDS
```

7. Step 5 — Build and start the application

Now you turn the project's source code into a single file and hand it to Payara.

1. Open a terminal in the project's main folder (the one containing `pom.xml`).
2. Run:

   ```
   mvn clean install
   ```

   This takes a minute or two the first time. It downloads what it needs, then builds the application. Wait for **BUILD SUCCESS**.
3. The finished file appears at:

   ```
   globaltrade-ear/target/globaltrade-ear.ear
   ```

4. Install it into Payara. Either use the control panel — **Applications → Deploy**, choose the file, click OK — or run this command:

   ```
   asadmin deploy --force=true globaltrade-ear/target/globaltrade-ear.ear
   ```

> **If the build stops with test failures**, the application file won't be created. You can skip the tests with `mvn clean install -DskipTests` if you just need it running.

8. Step 6 — Open it and log in

Open your browser and go to:

```
http://localhost:8080/globaltrade
```

You'll land on the login page. Sign in with any of the accounts in the next section — try `admin@globaltrade.com` with the password `123456` to start.

You don't have to pick a role or remember any other web address. The system recognises who you are and takes you to the right screen automatically. The menu at the top of the page changes to show only what you're allowed to use.

To sign out, click **Log out** in the top menu.

9. Login accounts

The starter data includes six ready-made accounts. **Every one of them uses the password `123456`.**

Email address	Password	Type of user	Name
`ops@abc-electronics.com`	`123456`	Customer	ABC Electronics
`shipping@globaltextiles.com`	`123456`	Customer	Global Textiles Ltd
`priya@techimports.com`	`123456`	Customer	Priya Fernando
`nadia@globaltrade.com`	`123456`	Coordinator	Nadia Perera
`kasun@globaltrade.com`	`123456`	Coordinator	Kasun Silva
`admin@globaltrade.com`	`123456`	Administrator	Admin User

> **These are demonstration accounts only.** The password is deliberately simple so the system is easy to try out. It is not what you would use in real life.

Creating your own account

Click **Register** on the login page to make a new account. New accounts are always created as **customers**. If you want a new coordinator or administrator, register the account first, then sign in as an administrator and change its type on the Users page.

10. What each type of user can do

The three types are separate, not ranked. An administrator is not “a customer with extra powers” — they simply do a different job, and the system won't let them into the customer screens at all.

Customer

Where: `/globaltrade/customer/dashboard` — shown in the menu as **My Shipments**

A customer can:

- See a list of their own shipments, with the route, current status, expected arrival date, estimated cost and number of containers.
- Book a new shipment by picking a starting port, a destination port, and how many containers they need.

Customers only ever see their own shipments. One customer cannot see another customer's business, even by trying different web addresses.

A few rules apply when booking:

- The starting and destination ports have to be different.
- At least one container is needed, up to twenty.
- Enough containers must actually be free. If not, the booking is refused and nothing is set aside.
- The cost works out at **$1,000 per container**.
- New bookings start as *Pending* with no arrival date yet.

Coordinator

Where: `/globaltrade/coordinator/dashboard` — shown in the menu as **Active Shipments**

A coordinator can:

- See every shipment in the system that hasn't been delivered yet, for all customers.
- Move a shipment on to its next stage.

Shipments move through set stages, and only certain moves make sense:

Current stage	Can move to
Pending	Confirmed, or Delayed
Confirmed	In Transit, or Delayed
In Transit	Delivered, or Delayed
Delayed	In Transit, or Delivered
Delivered	Nothing — the journey is finished

Anything else is refused with a short explanation. There's also one extra condition: to set a shipment *In Transit*, there has to be a ship docked at its starting port. If there isn't, the system says so and leaves the shipment alone until a ship is available.

> **Want to see that in action?** The starter data has no ship at Jebel Ali. Book a shipment starting there as a customer, then try to send it In Transit as a coordinator — you'll get the message. Add a ship at Jebel Ali from the admin Ships page and try again, and it works.

Administrator

Administrators have five screens, all listed in the top menu:

Menu item	Web address	What it's for
Containers	`/admin/containers`	See all containers, add new ones, change their status
Ships	`/admin/ships`	See all ships, add new ones, change status and location
Users	`/admin/users`	See everyone with an account and change what type of user they are
Audit Log	`/admin/auditLog`	A record of everything that has happened, newest first
Performance	`/admin/performance`	How fast the system is running

A few things worth knowing:

- Container numbers have to be unique. Adding one that already exists is politely refused.
- In the Audit Log, anything the system did on its own — rather than a person doing it — shows the user as **SYSTEM**.
- Administrators cannot book shipments or change shipment stages. Those belong to customers and coordinators.

The system also works on its own

Two things happen automatically in the background, without anyone clicking anything:

- Every fifteen minutes, the system checks for shipments that have reached their arrival date, marks them delivered, frees up their containers and updates the ship's location.
- Half an hour after a booking is made, the system checks whether it has moved on from Pending. If it hasn't, it notes an alert in the Audit Log.

Both of these show up in the Audit Log as SYSTEM entries.

11. A five-minute tour

If you want to see the whole system working end to end, do this in order. Log out between each step, or use a different browser window for each person.

1. **Sign in as** `ops@abc-electronics.com` (a customer). Book a shipment from Colombo to Rotterdam with 2 containers. It appears in your list as Pending, costing $2,000.
2. **Sign in as** `nadia@globaltrade.com` (a coordinator). Find that shipment in the list, move it to Confirmed, then to In Transit. A ship is assigned to it automatically.
3. **Sign in as** `admin@globaltrade.com` (an administrator). Open the **Audit Log** and you'll see both actions recorded against the people who did them. Then open **Performance** to see how long each one took.

> **Use a separate browser window for each person**, or log out properly between them. If you don't, the system may still think you're the previous user and show you a “not allowed” message that looks like a fault but isn't.

12. If something doesn't work

The Ping button fails when setting up the database connection

Two likely causes. Either the MySQL connector file wasn't copied into Payara's `lib` folder, or Payara wasn't restarted afterwards — go back to step 4a. Otherwise, check the username, password and database name you typed into the properties list. Try logging into MySQL with the same details from a terminal to confirm they're right.

The application won't install, or fails straight after installing

Almost always the connection name. It has to be exactly `jdbc/GlobalTradeDS`. Check it under **Resources → JDBC → JDBC Resources**, and remember it's case-sensitive.

The browser says the page can't be found

Check Payara is actually running — open `http://localhost:4848`. If that doesn't load, start it with `asadmin start-domain`. Also check the web address has no trailing extras: it's just `http://localhost:8080/globaltrade`.

Login says the email or password is wrong, but it should be right

The starter data may not have loaded properly. Log into MySQL and run:

```
USE globaltrade;
SELECT email FROM users;
```

If nothing comes back, run the setup script from step 2 again. Also double-check the password: it is `123456` — six characters.

A page says you're not allowed, even though you should be

You're probably still signed in as someone else. Go to `http://localhost:8080/globaltrade/logout` and sign in again.

Something else went wrong

Payara keeps a detailed record of everything it does. If a message on screen isn't helpful, the real reason is usually in this file:

```
payara6/glassfish/domains/domain1/logs/server.log
```

Open it and look at the last few lines — the most recent entries relate to whatever just happened.